

BLACKPOOL, United Kingdom

WMO: 033180

Lat: 53.774N

Lon: 3.039W

Elev: 34

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.0	28.8	21.4	16.0	30.5	24.0	18.2	32.0	63.3	44.8	58.7	44.5	5.9	100	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	74.6	63.0	70.6	61.9	68.1	60.9	65.0	70.8	63.5	68.4	62.3	66.4	12.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	85.9	67.0	61.6	82.1	65.5	60.4	78.6	64.4	30.0	71.0	28.9	68.5	28.0	66.7	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 51.1	(b) 42.7	(c) 31.4	(e) 23.4	(f) 83.3	(g) 1.7	(h) 3.8	(i) 22.2	(j) 86.1	(k) 21.2	(l) 88.3	(m) 20.2	(n) 90.4	(o) 19.0	(p) 93.2		
(a) 51.1	(b) 42.7	(c) 31.4	(e) 22.6	(f) 69.5	(g) 1.9	(h) 2.7	(i) 21.3	(j) 71.5	(k) 20.1	(l) 73.0	(m) 19.1	(n) 74.5	(o) 17.7	(p) 76.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	40.9	41.2	43.1	47.6	52.8	57.8	61.3	60.7	57.2	52.2	45.4	42.7		
	DBStd	8.74	5.02	4.71	4.73	4.88	5.07	3.94	3.62	2.97	3.80	4.65	5.56	6.12		
	HDD50	1289	282	248	217	102	24	1	0	0	2	30	154	229		
	HDD65	5388	747	667	678	522	380	220	126	135	235	397	589	692		
	CDD50	1396	0	0	4	30	111	234	351	333	218	98	15	2		
	CDD65	24	0	0	0	0	2	4	12	4	2	0	0	0		
	CDH74	147	0	0	0	0	20	37	78	7	4	1	0	0		
	CDH80	21	0	0	0	0	1	6	13	1	0	0	0	0		
	WSAvg	13.6	15.7	15.0	13.0	12.2	12.8	11.6	11.7	13.3	13.0	13.6	14.6	17.1		
	PrecAvg	37.7	3.5	2.8	2.3	2.1	2.4	2.4	2.6	3.4	3.5	4.3	4.1	4.1		
	PrecMax	49.3	5.9	5.9	5.2	3.6	4.8	5.7	5.6	7.4	6.9	9.6	7.7	9.9		
(15) Precipitation	PrecMin	28.7	0.3	0.5	0.8	0.6	0.6	1.1	1.0	0.6	0.4	1.1	2.1	1.1		
	PrecStd	5.6	1.4	1.3	1.2	0.9	1.1	1.2	0.9	1.4	1.5	1.7	1.5	2.0		
	0.4%	DB	52.6	55.5	60.9	69.4	77.1	78.9	81.8	74.1	72.4	66.0	58.1	55.4		
	0.4%	MCWB	50.5	49.2	51.8	57.5	61.8	63.1	66.1	63.7	63.1	60.0	56.1	53.0		
	2%	DB	50.0	50.7	54.9	63.9	69.9	71.6	75.1	70.1	68.3	62.6	56.4	52.9		
(23) Mean Coincident Wet Bulb Temperatures	2%	MCWB	48.4	48.0	50.6	53.0	59.9	61.4	63.7	62.4	61.2	58.5	54.4	50.8		
	5%	DB	48.5	49.0	51.8	58.6	64.3	67.5	70.7	67.6	65.6	60.6	54.7	51.4		
	5%	MCWB	46.8	47.0	48.1	51.3	56.3	59.7	62.8	61.5	59.8	57.6	52.6	49.6		
	10%	DB	47.4	47.4	49.6	54.9	60.4	64.8	67.9	65.6	63.3	58.8	53.0	50.0		
	10%	MCWB	45.7	45.6	46.8	50.1	54.7	58.4	61.5	60.3	58.6	55.8	50.9	48.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.0	50.1	54.1	58.0	63.7	66.0	68.7	65.9	65.6	61.6	56.7	53.4		
	0.4%	MCDB	52.0	52.2	58.1	67.9	74.4	75.1	79.5	70.2	70.9	64.3	57.7	54.8		
	2%	WB	48.6	48.6	51.0	54.9	60.4	62.4	65.5	64.0	62.8	59.7	54.9	51.4		
	2%	MCDB	49.5	50.3	54.4	61.6	68.3	68.7	71.8	67.9	66.1	62.0	56.0	52.5		
	5%	WB	47.4	47.4	49.0	52.6	57.9	60.8	63.8	62.8	61.0	58.3	53.1	50.0		
(33) Mean Daily Temperature Range	5%	MCDB	48.3	48.9	51.2	56.8	63.0	66.5	69.3	66.2	63.8	60.4	54.4	51.2		
	10%	WB	46.2	46.2	47.5	50.7	55.7	59.3	62.4	61.8	59.5	56.4	51.2	48.6		
	10%	MCDB	47.2	47.5	49.5	54.6	59.6	64.0	66.8	65.0	62.5	58.4	52.8	49.9		
	5% DB	MDBR	7.2	8.4	10.8	12.1	12.2	11.8	11.0	9.7	10.8	9.9	8.6	8.1		
	5% WB	MCDBR	6.7	8.8	14.0	19.0	19.3	18.1	17.7	13.1	15.1	11.6	7.9	8.6		
(40) Clear-Sky Solar Irradiance	taud	2.307	2.375	2.316	2.283	2.318	2.353	2.355	2.407	2.443	2.442	2.376	2.282			
	Ebn at Noon	197	239	255	266	271	272	269	267	255	233	198	175			
	Edh at Noon	19	24	32	37	38	37	36	32	28	23	18	16			
	RadAvg	192	409	773	1207	1546	1601	1525	1246	884	485	237	146			
	RadStd	21	45	63	109	115	148	151	82	68	45	25	14			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page